

Lab 01

Software Verification and Validation Lab

SWE 4806



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Introduction

Alloy is a textual modelling language aimed at expressing the structural and behavioural properties of software systems.

Installation Guide of Alloy Analyzer (AA)

Alloy is a self-contained executable that includes the Kodkod model finder, various SAT solvers, and the standard Alloy library.

1. Download Alloy from the official source [GithubLink](#).
2. Alloy requires a Java Runtime Environment (JRE). Download and install Java.
3. Run Alloy Analyzer:
 - Double-click the `.jar` file, **or**
 - Run the following command in the terminal:

```
java -jar alloy4.jar
```

Verification of Installation

To verify that Alloy Analyzer is installed correctly:

1. Click the **File** menu and select **Open Sample Models**.
2. Open the file:

```
examples/toys/ceilingsAndFloors.als
```

3. Click the **Execute** icon.

If Alloy Analyzer is successfully installed, it will display graphical output.